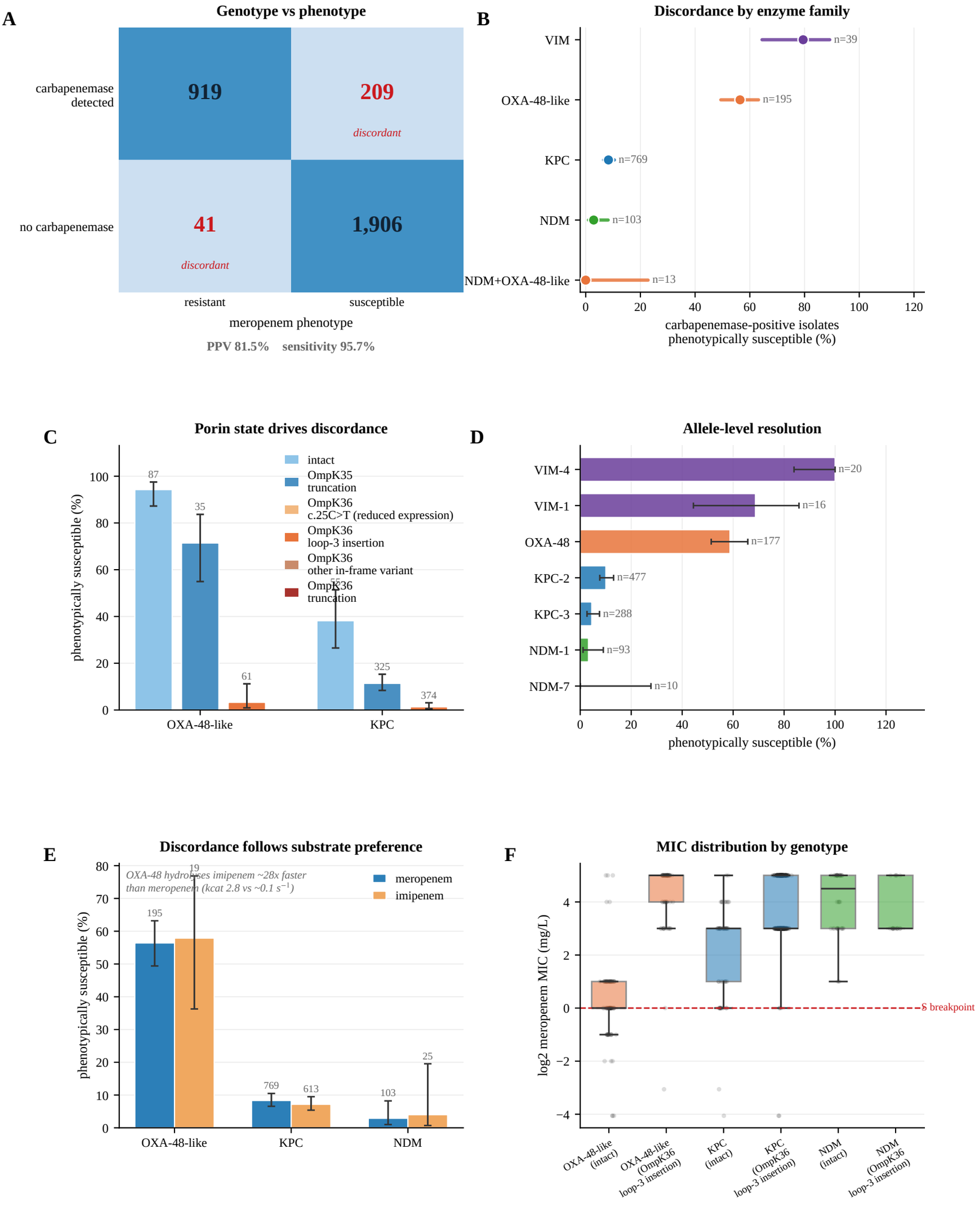
A
Genotype vs phenotype
carbapenemase detected
no carbapenemase
919
209
discordant
41
discordant
1,906
resistant
susceptible
meropenem phenotype
PPV 81.5%
sensitivity 95.7%
B
Discordance by enzyme family
VIM
OXA-48-like
KPC
NDM
NDM+OXA-48-like
n=39
n=195
n=769
n=103
n=13
carbapenemase-positive isolates phenotypically susceptible (%)
C
Porin state drives discordance
phenotypically susceptible (%)
intact
OmpK35 truncation
OmpK36 c.25C>T (reduced expression)
OmpK36 loop-3 insertion
OmpK36 other in-frame variant
OmpK36 truncation
87
35
61
55
325
374
OXA-48-like
KPC
D
Allele-level resolution
VIM-4
VIM-1
OXA-48
KPC-2
KPC-3
NDM-1
NDM-7
n=20
n=16
n=177
n=477
n=288
n=93
n=10
phenotypically susceptible (%)
E
Discordance follows substrate preference
OXA-48 hydrolyses imipenem ~28x faster than meropenem (kcat 2.8 vs ~0.1 s⁻¹)
meropenem
imipenem
phenotypically susceptible (%)
195
19
769
613
103
25
OXA-48-like
KPC
NDM
F
MIC distribution by genotype
log2 meropenem MIC (mg/L)
S breakpoint
OXA-48-like (intact)
OXA-48-like (OmpK36 loop-3 insertion)
KPC (intact)
KPC (OmpK36 loop-3 insertion)
NDM (intact)
NDM (OmpK36 loop-3 insertion)